

- 21** 2-Chlorobutane reacts with the :CH_3^- nucleophile to produce 2-methylbutane via nucleophilic substitution.

Before the reaction, the sample of 2-chlorobutane rotates plane-polarised light. After the reaction, the reaction mixture containing 2-methylbutane does not rotate plane-polarised light.

What can be concluded from the information given?

- 1 The reaction proceeds via $\text{S}_{\text{N}}1$.
- 2 The reaction proceeds via $\text{S}_{\text{N}}2$.
- 3 The rate of reaction is dependent on the concentration of 2-chlorobutane.

A 1 and 3 only **B** 2 and 3 only **C** 1 only **D** 3 only